| Paper ID | Citation | Study type | Impact factor(s) - RQ1 | Metric / measurement - RQ2 | Practice(s) - RQ3 | Lifecycle stage | Reported evidence |
|---|---|---|---|---|---|---|---|
| **P5** | Dash 2025. Green | empirical | Energy (training & inference) | Real-time energy monitoring; % energy rec | Multi-objective (Pareto) energy-vs-accuracy optimi | Training + inference + operation | Empirical (30.6% energy cut, 0.7% accuracy loss) |
| **P7** | Yarally, Cruz, Feito | empirical | Energy | Energy measurement during DL training | Energy-efficient training practices (HP/algorithm/ha | Training | Empirical |
| **P9** | Schwartz, Dodge, | position | Compute / efficiency (energy p | FLOPs (HW-agnostic); params; dataset siz | Treat efficiency as a first-class evaluation metric; r | Cross-cutting (research practice) | Position/argument |
| **P10** | Alizadeh & Castor | empirical | Energy | Energy of DL models across runtime infras | Runtime/infrastructure choice for inference efficien | Inference/serving (runtime) | Empirical |
| **P11** | Georgiou, Kechagi | empirical | Energy, cost | Energy cost of DL frameworks (TF vs PyTo | Framework choice as an efficiency practice | Training + inference | Empirical (ICSE) |
| **P15** | Sharma & Mathur 2 | framework | Energy | Energy profiling | Energy profiling, adaptive compression, fine-tuning | Model design + training + inference | Conference (new) |
| **P20** | Ferro, Silva, de Pa | empirical | Energy | Performance & energy efficiency of ML alg | Algorithm selection for energy efficiency (decision- | Model design / training | Case study |
| **P21** | Verdecchia, Cruz, | empirical | Energy, carbon | Energy measurement vs dataset reduction | Data-centric practices: reduce dataset size / impro | Data | Empirical (ICT4S) |
| **P23** | Kaswan, Kumar, R | position | Carbon footprint | Discussed, not measured | Model compression, transfer learning, low-power a | Cross-cutting | Perspective (IEEE) |
| **P25** | Acmali, Ortakci, Se | empirical | Energy (electricity of CNNs) | FLOPs, parameter count | Weight-level pruning to remove redundant parame | Model design (compression) | Empirical (8% params, 421M FLOPs cut, accuracy kept) |
| **P26** | Aggarwal, Sharma | framework | Energy, carbon | Proposed energy-aware AI evaluation fram | Model compression, efficient design, intelligent trai | Model design + training | Proposed framework (conceptual) |
| **P28** | Krishnaram 2025. | empirical | Energy (AI workloads) | Workload energy efficiency in data centres | Optimise energy efficiency of AI workloads (schedu | Operation/infrastructure | Conference (IEEE GreenTech) |
| **P29** | Clemm, Stobbe, W | analysis | Energy, carbon (status) | Status review of measurement | Research directions + green-AI practices across lif | Cross-cutting | Overview/agenda (EGG) |
| **P30** | Fourati 2025. Posit | position | Environmental cost of DL | - | Call for sustainable-AI practices (general) | Cross-cutting | Position (IJCNN) |
| **S1** | Geissler et al. 2024 | empirical | Energy (training); life-cycle-aw | HPC energy monitoring; energy-to-reach-t | Tune training hyperparameters (learning rate, batc | Training (hyperparameter configuration) | Empirical (multiple HP setups x 3 hardware systems) |
| **S2** | Strubell, Ganesh, M | empirical | Energy, CO2e, financial cost | kWh (power×time×PUE); CO2e via region | Report training time & HP sensitivity; prefer efficien | Training, NAS/HP search | Empirical estimates for NLP models |
| **S3** | Bender, Gebru, Mc | position | Environmental cost of large LM | - | Consider env cost; favour smaller/curated models | Data + training | Position (FAccT) |
| **S4** | Wu et al. 2022. Su | analysis | Energy, operational + embodie | At-scale footprint characterization (Meta) | Efficient models/data/hardware; operational practic | Cross-cutting (incl. inference, operation) | Industry data (Meta) |
| **S5** | Yang, Chen, Sze 2 | empirical | Energy | Energy modelling for CNNs | Energy-aware pruning of CNNs | Model design (pruning) | Empirical (CVPR) |
| **S6** | Gutiérrez, Moraga, | empirical | Energy | Energy impact measurement of optimisatio | Compare energy impact of different ML model opti | Model design / training | Empirical (ICT4S) |
| **S7** | Li, Chen, Becchi, Z | empirical | Energy (training + inference of | Energy consumption measured on CPUs v | Hardware selection (CPU vs GPU) for energy effic | Training + inference (hardware) | Empirical (CNNs on CPU/GPU, IEEE BDCloud/SustainCom) |
| **S8** | Henderson, Hu, Ro | tool | Energy, carbon | experiment-impact-tracker; kWh, CO2e, re | Adopt systematic energy/carbon reporting; carbon- | Training + measurement | Tool + framework applied to RL/ML |
| **S9** | Lacoste, Luccioni, | tool | Carbon | ML CO2 Impact calculator: emissions = f(h | Estimate & report emissions; choose low-carbon re | Training + cross-cutting | Calculator |
| **S10** | Anthony, Kanding, | tool | Energy, carbon | Carbontracker; predicts+tracks kWh & CO | Track/report; predict before training; stop unpromis | Training + monitoring/operation | Tool, demonstrated on DL training |
| **S11** | Desislavov, Martíne | analysis | Energy (inference) | Compute & energy trends in DL inference | Awareness of inference cost; efficiency gains | Inference/serving | Trend analysis |
| **S12** | Jacques, Alizadeh, | empirical | Energy (battery; on-device infe | Battery usage + inference performance me | Framework & model choice for energy-efficient on- | Inference/serving (mobile/edge) | Empirical (3 models x 3 frameworks x iOS devices, MOBILESoft 2024) |
| **S13** | Canziani, Paszke, | analysis (empirical | Power consumption, energy (re | accuracy, memory footprint, parameters, o | Use these metrics to design/select efficient DNNs; | Inference/serving (model design/selectio | Empirical analysis of ImageNet DNN architectures |
| **S14** | Aquino-Brítez et al. | empirical (propose | Energy (training + inference of | New standardized 'energy consumption inc | Use a standardized energy index to compare/selec | Model design/selection + training + infer | Empirical comparative analysis (MDPI 2025) |
| **S15** | Tabbakh, Al Amin, | framework | Energy, environmental impact | Overview (no new metric); reviews reportin | Model optimization methods; efficient algorithms; e | Model design + hardware + operation (c | Framework + case studies (NLP, CV) |
| **S16** | Van Wynsberghe 2 | conceptual | Sustainability of AI (concept) | - | Sustainable AI by design' principles | Cross-cutting | Conceptual (AI & Ethics) |
| **S17** | García-Martín, Lav | analysis | Energy | Methods/models to estimate ML energy co | Energy estimation & modelling; identify energy hot | Cross-cutting (measurement) | Analysis of estimation methods (JPDC 2019) |
| **S18** | Radovanović et al. | empirical (system) | Carbon (operational; datacente | Google Carbon-Intelligent Compute Manag | Carbon-aware temporal scheduling: shift/delay tem | Operation/infrastructure (scheduling) | Empirical (Google fleet; IEEE Trans. Power Systems) |
| **S19** | Kaack, Donti, Strub | analysis / framewo | GHG emissions of ML: (1) com | Systematic framework describing ML->GH | Policy levers to shape ML climate effects; measure | Cross-cutting (computing footprint + app | Framework/perspective (Nature Climate Change 2022) |
| **S20** | Patterson et al. 202 | analysis | Operational energy, CO2e | kWh, tCO2e; the '4 Ms' (Model, Machine, M | Sparse/efficient models (MoE); efficient accelerato | Training + operation/infrastructure | Measured estimates (T5, GPT-3, etc.) |
| **S21** | Patterson et al. 202 | analysis | Carbon | CO2e projections | Efficient models, processors, datacentres ('4Ms') | Training + operation | Analysis/projection |
| **S22** | Dodge et al. 2022. | empirical | Carbon | Measured carbon intensity of AI in cloud; s | Carbon-aware region/time selection; cloud instanc | Training + operation (cloud) | Empirical (FAccT) |
| **S23** | Luccioni, Viguier, L | empirical | Carbon (training + embodied + | Full-lifecycle CO2e accounting of a 176B L | Lifecycle carbon accounting; transparency; efficien | Training + operation + embodied | Empirical (JMLR) |
| **S24** | Wang, Na, Strubell | empirical | Energy, carbon (fine-tuning; vs | Empirical energy/carbon of fine-tuning acro | Recommendations to improve fine-tuning energy e | Training (fine-tuning) + measurement | Empirical (EMNLP Findings 2023) |
| **S25** | Prakash, Stewart, I | analysis (LCA) | Carbon + e-waste; operational | Life-cycle analysis (LCA) of TinyML at MCU | Assess whole life cycle (operational + embodied) t | Inference/serving (edge/MCU) + embod | Case studies + LCA (CACM 2023) |
| **S29** | Prakash, Stewart, I | analysis (LCA) | Carbon + e-waste; operational | Life-cycle analysis (LCA) of TinyML at MCU | Assess whole life cycle (operational + embodied) t | Inference/serving (edge/MCU) + embod | Case studies + LCA (CACM 2023) |